# Dense Passage Retrieval via Dual-Encoder Vector Representations (Karpukhin et al. 2020)

*Dense dual-encoder embeddings trained with in-batch negatives outperform traditional sparse lexical BM25 systems by 9–19% absolute top-20 passage retrieval accuracy in open-domain question answering.*

**GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED**

## WHAT WE KNOW (Narrowest Supported Conclusion)

Representing questions and candidate passages as continuous dense vectors via dual BERT encoders and indexing them via maximum inner product search (MIPS) dramatically outperforms traditional sparse inverted-index algorithms (BM25/TF-IDF) on semantic and paraphrased query retrieval.

## EVIDENCE AND CAUSAL ASSESSMENT

Controlled comparisons against BM25 holding downstream reader models constant isolate the dual-encoder dense vector space as the causal mechanism capturing synonyms, paraphrases, and semantic context beyond exact lexical token matches.

## EFFECT SIZE AND PERSISTENCE

On Natural Questions, DPR achieved 78.4% top-20 retrieval accuracy compared to 59.1% for BM25. Combining dense representations with sparse BM25 scores (hybrid retrieval) yielded further performance gains across all benchmarks.

## BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as: robust (<20%).

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## WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Dense retrieval does not guarantee semantic relevance across out-of-domain technical taxonomies without domain-specific continuous pre-training or fine-tuning on hard negatives.

## WHAT ACTION IS REASONABLY JUSTIFIED

Standardize on hybrid retrieval pipelines combining dense dual-encoder embeddings (e.g., DPR, BGE, or modern text embeddings) with sparse lexical search (BM25) and approximate nearest neighbor (ANN) vector indices.

## VERIFIED SOURCES & DOI IDENTIFIERS

- Karpukhin, V., Oğuz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., & Yih, W.-t. (2020). Dense Passage Retrieval for Open-Domain Question Answering. arXiv preprint arXiv:2004.04906. https://arxiv.org/abs/2004.04906